Boat Puck — full system breakdown (for Robby)

Audience: engineer / developer who must understand the whole race system end-to-end.
Date: 2026-09-04 · **FX for guide prices: R16 / $1** (always show both — see `PRICE_RULE.md`).
Product goal: cheaper Vakaros-class fleet race network (cm RTK + LoRa + start/line/OCS), not a Sailmon clone.

Locked V1 hardware bet: OTW/Anzewei **WT-43-RK-LoRa** (boat/pin/mark) + **WT-43-BK-LoRa** (committee base). **Unicore UM980** = benchmark / fallback. **UWB off every-boat BOM.**

Related digests:

Doc	What
`DEV_DIRECTION_2026-09_to_Shenzhen_2027-04.md`	Sep 2026 → Shenzhen Apr 2027 direction
`race-kit-roles-wt43-v1.md`	Who gets which box
`logic-check-wt43-v1.md`	Why WT-43 is V1 core
`accuracy-vs-racesense-pins.md`	Lipton R1–R10 OCS accuracy bar
`universal-puck.md`	Puck + optional screen/watch/tablet
`housing/README.md`	GoPro H9–13 shell packing
`puck-components-buy-list.md`	Alternate discrete UM980 path parts
`components-requirements.md`	Capability checklist vs Vakaros

Live Lipton references (same GPS, different UIs):

- Vakaros player: https://player.vakaros.com/watch/Lv9A35uOBSBRmGpHgXtH/J22
- SailingSA replay: https://sailingsa.co.za/regatta/2026-08-29-lipton-challenge-cup-dev
- Sailfish UI: https://sailingsa.co.za/tracking-dev2.html

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0. One picture of the whole system

```
                    ┌──────────────────────────────────┐
                    │ COMMITTEE BOAT (Race Control)     │
                    │ • WT-43-BK-LoRa = RTK BASE        │
                    │ • Survey antenna on pole          │
                    │ • Laptop/tablet = Race Control SW │
                    │ • Optional 4G for spectators only │
                    └────────────────┬─────────────────┘
                                     │ LoRa broadcast RTCM
                                     │ (no cellular on race path)
              ┌──────────────────────┼──────────────────────┐
              ▼                      ▼                      ▼
    ┌─────────────┐        ┌─────────────┐        ┌─────────────┐
    │ START PIN   │        │ MARK packs  │        │ BOAT PUCKS  │
    │ WT-43 rover │        │ WT-43 rover │        │ WT-43 rover │
    │ float/clip  │        │ float/clip  │        │ + nRF54L15  │
    │             │        │ each mark   │        │ + IMU+batt  │
    │             │        │             │        │ + GoPro case│
    └──────┬──────┘        └──────┬──────┘        └──────┬──────┘
           │ positions            │ positions            │ pos + OCS
           └──────────────────────┴──────── LoRa uplink ─┘
                                     │
                                     ▼
                            Race Control UI
                            (line, OCS, finish, tracks)

  Boat Puck ──BLE──► watch / phone / optional Screen housing
  (optional experiment) Bow Tag nRF54L15 ↔ Puck Channel Sounding
```

Hard rules

1. **One RTK base** for the whole venue (committee). Everything else is a **rover**.
2. **No cellular on the race-critical path** (RTCM, gun, OCS, finish). Optional 4G = media/spectator only.
3. **Live priority:** start/OCS → finish → mark roundings. Full track may **store + backfill** between events.
4. Marks and boats are **radios/relays**, not GNSS-only dumb sensors (Lipton dropped points upwind to M1).

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1. What problem we are solving (OCS bar)

From Lipton Challenge Cup J22 R1–R10 (Vakaros GPS):

Fact	Implication
Many OCS/clear calls sit in **~0.1–0.5 m** at the gun	Sub-metre DGNSS is **marginal**; need **cm RTK**
Example R1: KYC **+0.21 m** OCS vs GLYC **−0.19 m** clear	Decision gap **~0.4 m**
RC→M1 ~**1.5–2.3 km**; boats ~**2–2.5 km** from RC	LoRa design margin **≥3 km** usable
Worst dropouts **upwind to M1**, not always max range	Need **relays** + **on-device store/backfill**

Full write-up: `accuracy-vs-racesense-pins.md`.

OCS math (same class as RaceSense):

1. Live line = **committee end** + **start pin** (both RTK).
2. Boat reports antenna position; firmware applies **puck→bow lever arm** using heading/IMU.
3. At gun epoch **T=0** (GNSS-synced), signed horizontal distance of **bow** to line.
4. Publish confidence: **definite OCS / clear / grey zone** (do not pretend datasheet 1 cm = finished OCS 1 cm).

Error budget to **measure**: RTK fix · line ends · timing · bow geometry · heading.

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2. Roles — what sits where

2.1 Committee boat (Race Control)

Item	What	Why
WT-43-BK-LoRa	RTK **base**	Broadcasts RTCM over integrated LoRa to all rovers
Survey / patch antenna on pole	Better base antenna	On-module patch is weak for *base*; pole clears metal
12 V / large battery box	All-day power	Base must not die mid-race
Laptop or tablet	Race Control host	Start sequence, line, OCS list, finish, scoring, maps
USB/UART bridge (Pi or USB-serial)	Host ↔ base / hub	Pipe status, inject race messages, log
Optional 2nd WT-43 rover on bow	Clean “committee end” of line	Only if base pole ≠ the geometric line end
Optional 4G modem	Spectator uplink	**Never** required for OCS

V1 simplest layout: base antenna pole on committee **is** one end of the start line.

Guide price (committee core electronics):

Part	Guide	URL
WT-43-BK-LoRa	**~R512–900 ($32–56)** factory; Ali ~**R867 ($54)**	https://www.ontheway-tech.com/product/wt-43-bk-lora-module/
Survey patch + pole + box	**~R300–800 ($19–50)**	any marine/survey patch + PVC pole
Host laptop/tablet	bring existing	—
Committee core total	**~R850–1 700 ($53–106)** + host	

Factory contact pattern: `Lucaszhang@ontheway-tech.com` (confirm on reply).

2.2 Start pin (not committee end)

Item	What
WT-43-RK-LoRa	Rover — continuous cm position
Sealed **float / clip pack**	Survives dunk + pin clip
Battery (1–2 race days)	Same LoRa mesh as boats
No BLE watch stack required	Not a sailor UI device

Job: other end of start line → Race Control + every boat compute live line / OCS.

Guide price: ~R850–900 ($53–56) per pin pack (module-dominated).

Product page (rover): https://www.ontheway-tech.com/product/wt-43-rk-lora/

2.3 Marks (windward, leeward, gate, offset, …)

Item	What
1× WT-43-RK-LoRa rover pack per mark	Same brick as pin
Gate	**2 packs** (L + R)
Firmware role	Position report + **LoRa relay** for boats up the course

Job: live mark positions for tracking / bounds / optional hit-mark; **radio relay** so upwind boats don't go dark (Lipton lesson).

Not an RTK base. Never configure a mark as second base.

Guide price: ~R850–900 ($53–56) × N marks.

2.4 Finish

Layout	Hardware
Finish = start (common dinghy)	Reuse committee end + start pin — **no extra finish pin**
Separate finish line	Finish boat **rover** + finish pin **rover** (committee base stays at start)
Finish at a mark	That mark pack is the target

Rule: finish ends are always **rovers**. Only **one base** venue-wide.

2.5 Racing boat — the Puck

Primary sailor product: [`universal-puck.md`].

Layer	Part	Role
GNSS+LoRa brick	**WT-43-RK-LoRa**	RTK rover; receives RTCM; UART NMEA/positions out
App MCU	**nRF54L15** (prod) / Tag+DK (proto)	Fusion, OCS, LoRa protocol helper, BLE to UI, OTA, logging
Motion	**6–9 DOF IMU** (from Nordic Tag first, then BMI270-class)	Heel, heading, puck→bow lever
Power	**3000–5000 mAh** LiPo + charger	Full race day+
Shell	**GoPro H9–13 waterproof case**	Cheap, mount ecosystem, sky-friendly
Antenna	Module patch **or** case-top L1+L5 if needed	**Must prove** inside housing
Status	RGB LED	FIX / radio / OCS
Optional bow	**nRF54L15 Tag**	Bluetooth 6 Channel Sounding experiment
Optional UI	Phone / watch / Screen housing	BLE client only — **no LoRa on UI device**

Housing packing: cavity ≈ camera **71.8 × 50.8 × 33.6 mm**; lens tunnel = GPS well **~Ø32 × 5.5 mm**. See `housing/README.md`.

WT-43 size note: factory **43 × 43 × ~14 mm** (not 8.2 mm). Fits GoPro only with flat battery + tight sled — `logic-check-wt43-v1.md`.

2.6 Optional Screen / Atlas display (not V1 race-critical)

SKU	Role	Guide	URL
Waveshare ESP32-S3-RLCD-4.2	UI proto	**R400 ($24.99)**	https://www.waveshare.com/esp32-s3-rlcd-4.2.htm?sku=33507
Docs	—	—	https://docs.waveshare.com/ESP32-S3-RLCD-4.2

Doctrine: Screen is **BLE client of Puck**, not a second radio brain.

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3. Boat Puck — bill of materials (V1 primary path)

3.1 Primary V1 (WT-43 integrated)

#	Component	Qty/ boat	Guide price	Buy / datasheet
1	**WT-43-RK-LoRa** (rover)	1	**R512–900 ($32–56)**	https://www.ontheway-tech.com/product/wt-43-rk-lora/
2	**nRF54L15 DK** (bench) / module later	shared	DK ~**R1 600–2 400 ($100–150)** class via DigiKey/ Mouser	https://www.nordicsemi.com/Products/Development-hardware/nRF54L15-DK
3	**nRF54L15 Tag** (bow / IMU / CS)	0–1	~**R480–560 ($30–35)**	https://www.nordicsemi.com/Products/Development-hardware/nRF54L15-Tag
4	IMU (if not using Tag)	1	**R30–80 ($2–5)** BMI270-class	DigiKey / Mouser BMI270
5	LiPo 3–5 Ah + charge	1	**R80–200 ($5–12)**	any flat LiPo + TP4056-class
6	GoPro H9–13 waterproof case	1	**R80–100 ($5–6)**	AliExpress HERO9–13 protective housing
7	3D sled / insert	1	**R20–50 ($1–3)** print	design in `housing/`

#	Component	Qty/ boat	Guide price	Buy / datasheet
8	LED + passives + wiring	1	**R20–40 ($1–2.5)**	—
	Electronics+shell guide (WT-43 path)		**~R850–1 350 ($53–84)** early; volume lower	

Kill / prove gate for this path: moving **RTK FIX @ up to 20 Hz** with **LoRa RTCM** over water. Fail → UM980 path.

3.2 Fallback / benchmark (discrete UM980)

Use if WT-43 fails rate / FIX / antenna / LoRa on water.

#	Component	Guide	URL
GNSS	Unicore **UM980** board	**R960–2 080 ($60–130)** Ali · **R2 720 ($170)** trusted	https://en.unicore.com/products/um980/ · https://gnss.store/products/elt0223
MCU+BLE interim	Ebyte **E73** nRF52840	**R122 ($7.60)**	https://ebyteiot.com/products/2-4ghz-ble-mesh-small-smd-e73-2g4m08s1c-nordic-nrf52840-module-small-size-ble-5-0
LoRa discrete	Ebyte **E22-900M22S**	**R96 ($5.98)**	https://ebyteiot.com/products/sx1262-868mhz-module-electronic-components-22dbm-wireless-transceiver-lora-gfsk-iot-long-range-7km-ebyte-e22-900m22s-spi
Alt LoRa UART	**A39**-class	**R110–130 ($7–8)**	e.g. https://openelab.io/products/lora-a39-t900a30d1a (verify SA band)
Housing	same GoPro	**R80–100**	Ali
Puck total (UM980 path)		**~R1 900–3 500 ($120–220)**	`puck-components-buy-list.md`

3.3 Lab-only (do not put on every boat)

Part	Why	URL
Qorvo/Decawave **DWM3001C** UWB	Ranging reference only (~R700+/boat unjustified)	DigiKey DWM3001C
NXP MCXW72-LOC ×2	Independent BT Channel Sounding benchmark	NXP / DigiKey MCXW72 kits

3.4 How the Puck is assembled (proto)

1. Print/mill **sled** that locks WT-43 so the GNSS face looks out the **lens window**.
2. Place **battery flat** beside/under module (height budget is the tight constraint).
3. Wire **UART**: WT-43 TX/RX → nRF54L15 UART; share GND; 5 V or regulated rail per datasheet.
4. Wire **IMU** (SPI/I^{2}C) to nRF; or use Tag's onboard IMU for early tests.
5. LoRa antenna: keep clear of battery foil; if module antenna is blind in plastic, add **external L1+L5** through a sealed gland / case-top patch.
6. Close GoPro door; mount with **sky view** (transom / foredeck / pushpit — class rules permitting).
7. Calibrate **antenna → bow** offset in config (metres along centreline + lateral).

Production later: custom PCB + own mould — **not yet** (Shenzhen Apr 2027 evidence-first).

4. How the radio / data plane works

4.1 Downlink (critical): RTCM corrections

```
Committee WT-43 BASE  --LoRa broadcast RTCM⟶  every rover (boats, pin, marks)
```

- **Broadcast** — do **not** ACK every correction (fleet of 50–100 would melt the air).
- SA band / power compliance must be checked for 410–525 MHz class modules (WT-43 LoRa band on datasheet).
- Target usable range **≥3 km** over water at the air rate that still sustains FIX.

4.2 Uplink: positions, events, relays

```
Rovers --LoRa⟶ Race Control (and peer relays)
```

Suggested message classes:

Message	Who → whom	Priority
RTCM	Base → all	Continuous
Time / gun / start sequence	RC → all	Critical
Line ends / course / mark IDs	RC → all	High
Position + fix quality + boat ID	Boat → RC (+relay)	High near start/finish; backfill else
Pin / mark position	Pin/mark → RC + boats	High
OCS / penalty state	Boat ↔ RC	Critical at start
Finish cross event	Boat → RC	Critical
Track samples	Boat → RC	Store locally; uplink when quiet
Relay frames	Mark/boat → boat	As needed for coverage

Store + backfill (mandatory from Lipton):

- Puck logs **all** GNSS samples locally (flash).
- Live stream can drop; between roundings, **ACK/backfill** missing segments.
- “Jumps” in Vakaros trails were **missing samples**, not teleportation.

4.3 BLE (boat-local only)

```
Puck --BLE⟶ watch / phone / Screen housing
```

Carries: DTL to line, countdown, OCS flag, speed/heading, battery, config.
Does not carry RTCM for the fleet.

4.4 Optional BT6 Channel Sounding (experiment)

```
Bow Tag (nRF54L15 Tag)  ⟷  Puck (nRF54L15)   // 0.25–100 m ranging
```

Only interesting if sailing **95%** error ≤ ~5–10 cm. Else **RTK + geometry** stays authoritative for bow.

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5. How OCS works (software + geometry)

5.1 Inputs at gun

Input	Source
Line end A	Committee end (base antenna or bow rover)
Line end B	Start pin rover
Gun time T=0	Race Control start sequence, GNSS time sync over LoRa
Boat antenna (x,y)	Boat puck RTK at/near T=0 (interpolate if 10–20 Hz)
Heading / heel	IMU (+ mag if trusted)
Lever arm	Config: antenna → bow (and optional beam offset)

5.2 Algorithm (conceptual)

1. Build infinite line through A–B in **horizontal ENU / local metres**.
2. Transform antenna → **bow point** using heading + lever.
3. Signed distance of bow to line (course side = positive).
4. Classify vs thresholds (example policy — tune from data):
   - `> +grey` → **OCS**
   - `< -grey` → **clear**
   - else → **grey zone** (RC decision / individual recall UI)
5. Emit event to RC + LED + BLE alert on boat.

5.3 Why IMU still matters with cm RTK

- Antenna is rarely at the bow.
- Heel/leeway change the horizontal lever.
- Short GNSS blips need coasting.
- Heading quality drives OCS as much as centimetres of RTK on tight calls.

5.4 What must be developed (OCS stack)

Software piece	Where it runs
GNSS parse + time stamp	Puck MCU
Lever-arm + heading fusion	Puck MCU
Line geometry + signed distance	Puck **and/or** Race Control (prefer both: local alert + RC authority)
Gun scheduler / sync	Race Control → LoRa
OCS list UI + protest log	Race Control
Confidence / grey-zone policy	Both
Replay tools vs video	Shore / SailingSA tooling

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6. What software must be written (full list)

6.1 Embedded — Boat Puck firmware (nRF54L15 + WT-43)

Module	Responsibility
`gnss_uart`	Config WT-43 rate (target 10–20 Hz), parse NMEA/RTCM status, FIX quality

Module	Responsibility
`lora_link`	Join mesh/star, receive RTCM, send telemetry, relay, duty cycle
`time_sync`	Align to RC GNSS time / gun countdown
`imu_fusion`	Heel, pitch, yaw rate, heading estimate
`boat_model`	Antenna → bow/stern offsets per class
`ocs_engine`	Line distance, gun latch, flags
`finish_engine`	Crossing detection vs finish line/mark
`mark_events`	Rounding heuristics (optional V1.1)
`logger`	Ring buffer / flash; backfill protocol
`ble_gatt`	Sailor UI service (DTL, timer, OCS, battery, config)
`ota`	Firmware update (BLE or LoRa slow path)
`power`	Sleep between races, brownout, charge state
`led`	FIX / LOS / OCS / battery codes
`cs_ranging` (exp)	Channel Sounding to bow tag

6.2 Embedded — Pin / mark pack firmware

Same GNSS+LoRa brick, **stripped** UI:

- Rover FIX
- High-rate position broadcast near start (pin)
- Lower-rate + **relay** role (marks)
- Battery / health telemetry
- No BLE sailor stack required (optional BLE for setup phone)

6.3 Embedded — Committee base

- WT-43 **base mode**
- Continuous RTCM LoRa broadcast
- Health (satellites, age, TX duty) to Race Control host
- Optional: host injects race messages on same radio via MCU bridge

6.4 Race Control application (laptop/tablet)

Feature	Notes
Fleet map	Boats, pin, marks, line
Start sequence	5-4-1-gun; sync to GNSS time
Live line	From committee end + pin
OCS board	Auto list + grey zone + manual override
Individual recall / general recall	Flags to boats over LoRa
Finish / scoring	Auto cross + manual
Course editor	Mark order, gates, finish mode
Radio health	RSSI, FIX%, correction age
Logging / export	For protests + SailingSA ingest
Spectator export (optional)	4G only

Tech choice open (Electron / Flutter / web+local server). Must work **offline on water**.

6.5 Sailor apps

Client	Job
Phone (iOS/Android)	Setup, offsets, live DTL/timer/OCS
Watch	Eyes-up countdown / OCS vibe
Screen housing firmware	Digits + **loud** start/OCS beep

6.6 Shore / SailingSA integration (later)

- Ingest tracks into existing replay (`lipton-dev-trail-rN.json` class)
- Compare vs Vakaros for validation
- Not required for first on-water OCS proof

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7. Minimal kits and guide budgets

7.1 Bench prove (buy first)

Qty	Item	~R	~$
1	WT-43-BK-LoRa base	850–900	53–56
1	WT-43-RK-LoRa rover	850–900	53–56
2	nRF54L15 DK	~3 200–4 800	200–300
2	nRF54L15 Tag	~960–1 120	60–70
—	Wires, USB serial, bench PSU	200	12
	Bench subtotal	**~R6 000–8 000**	**~$375–500**

Pass gate: static then moving FIX with LoRa RTCM.

7.2 Two-boat OCS field kit

Qty	Item	Role	~R each
1	WT-43 base + pole	Committee	850–900 + ant
2	Boat Pucks	Rovers + MCU/IMU/case	850–1 350
1	Pin pack	Start pin	850–900
1	Host tablet	Race Control	existing
	Field electronics		**~R4 500–6 500 ($280–400)**

7.3 Club event kit (illustrative)

Role	Qty	~R
Base	1	900
Boat Pucks	20	20 × 1 100 ≈ 22 000
Start pin	1	900
Marks (W + gate + L)	4	3 600
Finish (if separate)	0–2	0–1 800
Rough fleet electronics		**~R28 000–30 000 ($1.7k–1.9k)**

Compare: Vakaros Atlas 2 ~**$1 249 (~R20 000) per boat** + HALO ~**$599**. Our boat puck target remains ≪ **Atlas**.

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8. Assembly matrix (what plugs to what)

From	To	Interface	Notes
WT-43 rover	nRF54L15	UART TTL	NMEA + status; baud up to 921600
WT-43 LoRa	peer WT-43 / hub	LoRa RF	RTCM / telemetry
IMU	nRF54L15	SPI or I^2C	≥100 Hz preferred
Battery	5 V rail / charger	power	WT-43 typ. 3.6–6 V (5 V)
nRF54L15	LED	GPIO	status
nRF54L15	Phone/watch	BLE	sailor UI
nRF54L15	Bow Tag	BLE CS	experiment
Base WT-43	Race Control host	USB-UART or BLE bridge	config + health
Race Control	Base radio	same	gun / course messages

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9. Development sequence (for Robby)

1. **Bench RTK:** 1 base + 1 rover, walk test, log FIX%, age of corrections.
2. **Rate:** lock 10 Hz then 20 Hz moving.
3. **MCU bring-up:** nRF54L15 reads GNSS UART; LED = FIX.
4. **OCS sandbox:** two fixed line ends + one moving rover; gun button; plot signed distance.
5. **Housing:** sled in GoPro; re-test antenna (may need external).
6. **Two boats + pin on water:** deliberate OCS vs clear vs video.
7. **Add windward mark pack:** measure uplink loss with/without relay.
8. **Store/backfill:** induce RF blackouts; prove log completeness.
9. **BT6 CS bow tag:** only after RTK OCS works.
10. **UM980 A/B:** if WT-43 fails gates.
11. **Shenzhen Apr 2027:** take working units + datasets + BOM asks 100/500/1000 — not “what tracker do you sell?”

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10. Clickable buy list (quick)

What	URL	Guide
WT-43-RK-LoRa (rover)	https://www.ontheway-tech.com/product/wt-43-rk-lora/	R512–900 ($32–56)
WT-43-BK-LoRa (base)	https://www.ontheway-tech.com/product/wt-43-bk-lora-module/	R512–900 ($32–56)
Unicore UM980	https://en.unicore.com/products/um980/	OEM
UM980 trusted board	https://gnss.store/products/elt0223	R2 720 ($170)
nRF54L15 DK	https://www.nordicsemi.com/Products/Development-hardware/nRF54L15-DK	~$100–150
nRF54L15 Tag	https://www.nordicsemi.com/Products/Development-hardware/nRF54L15-Tag	~$30–35

What	URL	Guide
Ebyte E73 nRF52840	https://ebyteiot.com/products/2-4ghz-ble-mesh-small-smd-e73-2g4m08s1c-nordic-nrf52840-module-small-size-ble-5-0	R122 ($7.60)
Ebyte E22 LoRa	https://ebyteiot.com/products/sx1262-868mhz-module-electronic-components-22dbm-wireless-transceiver-lora-gfsk-iot-long-range-7km-ebyte-e22-900m22s-spi	R96 ($5.98)
Waveshare 4.2" RLCD	https://www.waveshare.com/esp32-s3-rlcd-4.2.htm?sku=33507	R400 ($24.99)
GoPro H9–13 case	AliExpress “HERO9 HERO10 HERO11 HERO12 HERO13 protective housing”	R80–100 ($5–6)

Do not buy as Boat Puck: locked indoor iBeacon-only nRF54L tags (e.g. KKM “NRF54L Bluetooth 6.0 Beacon”) — wrong firmware class for Channel Sounding / open Nordic SDK. Use **Nordic’s** Tag/DK.

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11. Glossary

Term	Meaning
Base	Stationary RTK reference that emits corrections
Rover	Moving RTK receiver (boat, pin, mark)
RTCM	Correction messages for RTK
FIX	RTK fixed integer solution (cm-class)
OCS	On course side (over early) at start
DTL	Distance to line
Puck	Boat unit in GoPro shell
Pin pack	Sealed rover on start/finish pin
Mark pack	Sealed rover on a course mark (+ relay)
Race Control	Committee software + operator

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12. One-page “if you remember nothing else”

1. **Committee** = 1× WT-43 **base** + antenna pole + Race Control laptop.
2. **Pin / marks / finish ends** = WT-43 **rovers** in float packs.
3. **Every boat** = WT-43 rover + nRF54L15 + IMU + battery in **GoPro** case.
4. **LoRa** carries RTCM down and positions/events up; **BLE** is sailor UI only.
5. **OCS** = bow vs live RTK line at GNSS gun time, with grey zone.
6. **Write:** puck firmware, pin/mark firmware, Race Control app, sailor BLE apps, logger/backfill.
7. **Prove** moving LoRa-RTK before scaling; **UM980** if WT-43 dies; **no UWB** on every boat.
8. Prices above are **guide** at R16/$1 — re-quote before PO.

End of Robby breakdown. Keep this file as the single onboarding map; deep dives stay in the linked docs.